

1. Russell's Paradox considers the set of all sets that don't contain themselves (i.e. $\{x : x \notin x\}$), and gets into a contradiction.

Use Russell's Paradox to show that every set has a subset that is not an element of it.

Solution: Let A be a set. We want to show that A has a subset which is not an element of A .

We consider $R = \{x \in A : x \notin x\}$. Suppose $R \in A$. Then we get $R \in R \Leftrightarrow R \notin R$, which is a contradiction. Hence $R \notin A$.

2. Recall that a function $f : A \rightarrow B$ is a *surjection* iff every $b \in B$ gets "hit" by some $a \in A$, i.e. $f(a) = b$. Show that there is no surjection from a set X onto $\mathcal{P}(X)$.

Solution: Let $f : X \rightarrow \mathcal{P}(X)$ be a surjection. We derive a variant of Russell's Paradox. Consider the set $R = \{x \in X : x \notin f(x)\}$. Since f is a surjection, there is some $r \in X$ with $f(r) = R$. Plugging in r into our definition of R gives $r \in R \Leftrightarrow r \notin f(r)$. But since $f(r) = R$, we get $r \in R \Leftrightarrow r \notin R$, which is the same contradiction.

3. Recall that a function $f : A \rightarrow B$ is an *injection* iff whenever $f(x) = f(y)$, we get $x = y$. Prove that there is no injection from the powerset of a set X into X .

Solution: Let $f : \mathcal{P}(X) \rightarrow X$ be an injection. We will make a surjection that goes the other way, which by the last problem is impossible.

Pick some subset $E \subseteq X$ (e.g. $E = \emptyset$). For each $x \in X$ define $g(x)$ as follows. If there is an $A \subseteq X$ s.t. $f(A) = x$, then we define $g(x) = A$ (we know there can be at most one such A). Otherwise, we define $g(x) = E$. Since f is an injection defined everywhere on $\mathcal{P}(X)$, we know that for each $A \in \mathcal{P}(X)$ gets hit by $f(A)$, and hence, g is a surjection. Contradiction!

4. Conclude that for any set X , we have $|X| < |\mathcal{P}(X)|$.

Solution: It suffices to show that there is an injection from X into $\mathcal{P}(X)$, and no injection from $\mathcal{P}(X)$ into X .

An injection from X into $\mathcal{P}(X)$ is given by $x \mapsto \{x\}$. By problem 3 there is no injection going the other way.